

vtslides

Typst slides with View Transitions

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$$\int_0^\infty e^{-x^2} \mathrm{d}x = \frac{\sqrt{\pi}}{2}$$

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The quick brown fox

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cell

cell

cell

cell

cell

cell

cell

Overlap

Two marks, A and B, over five frames: overlapping, parting, crossing diagonally, passing through each other.

Overlap

AAAAAAA
BBBBBBBB

Overlap

BBBBBBBBBB

AAAAAAA

Overlap

BBBBBBBBBB

AAAAAAA

Overlap

AAAAAAAAAA

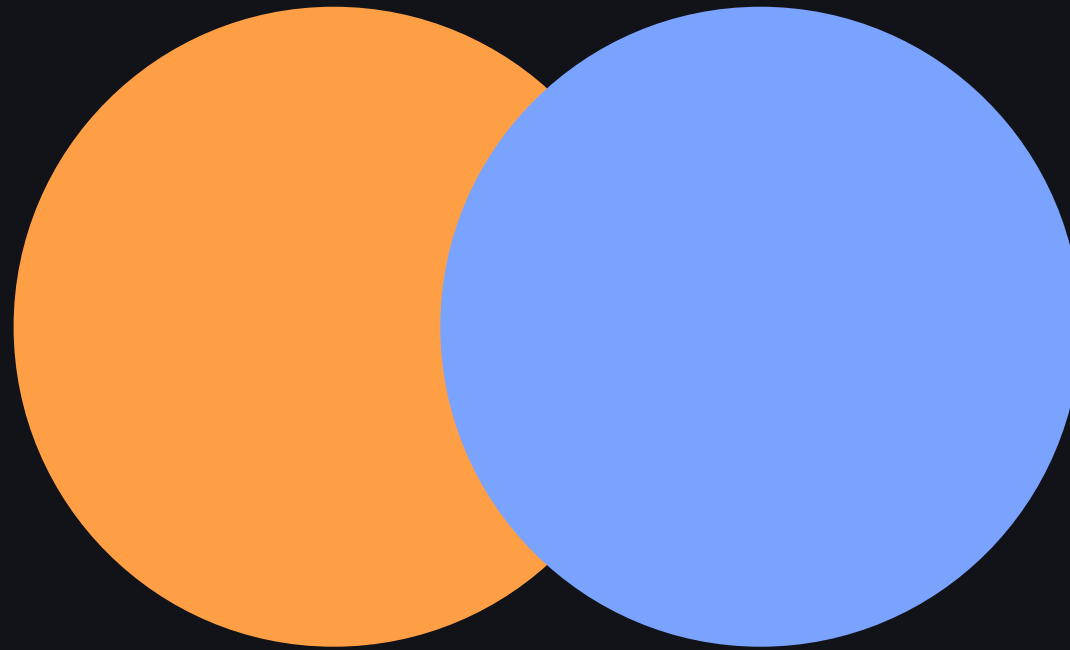
BBBBBBBBBB

Overlap

BBBBBBBBBB

AAAAAAAAAA

Discs



Discs

Discs

vtslides

```
mark("word", transition: "zoom")
```

Cauchy sequences

Definition 1 (Cauchy sequence). A sequence (a_n) is **Cauchy** if for every $\varepsilon > 0$ there is an N such that $|a_m - a_n| < \varepsilon$ whenever $m, n > N$.

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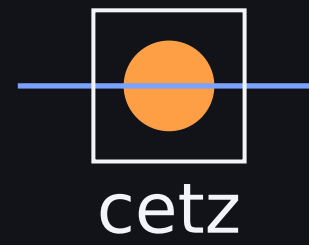
Theorem 2. *A real sequence converges if and only if it is Cauchy.*

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Definition 1 (Cauchy sequence). A sequence (a_n) is **Cauchy** if for every $\varepsilon > 0$ there is an N such that $|a_m - a_n| < \varepsilon$ whenever $m, n > N$.

Theorem 2. *A real sequence converges if and only if it is Cauchy.*

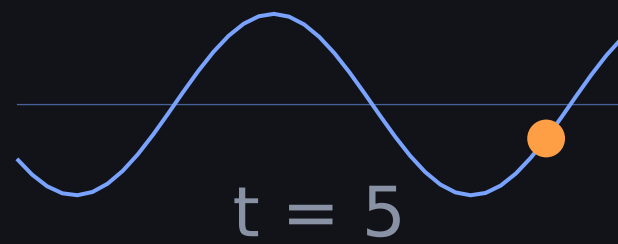
Proof. Convergent implies Cauchy: $|a_m - a_n| \leq |a_m - a| + |a - a_n|$. A Cauchy sequence is bounded, so by Bolzano–Weierstrass it has a convergent subsequence $a_{n_k} \rightarrow a$, and the Cauchy condition pulls the whole sequence to a . □





```
mark("fig")[#cetz.canvas(...)]
```


Element animation



```
mark("wave", ..range(0, 6).map(wave))
```

Continuous animation



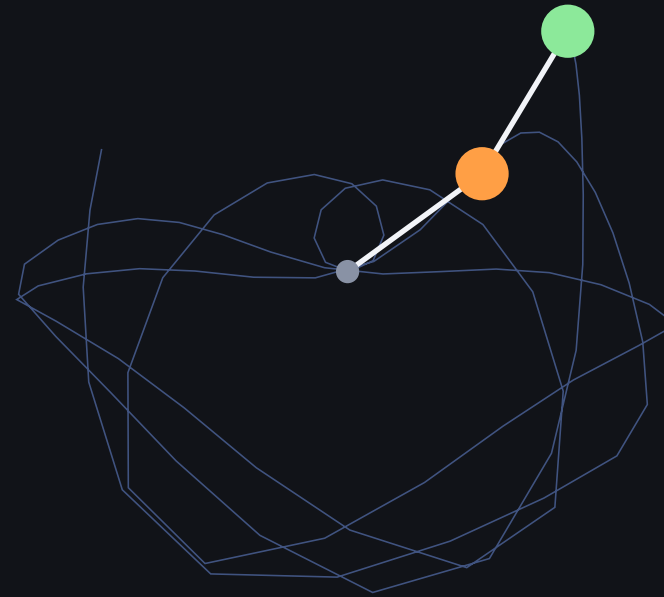
```
anim: (dot: (follow: "track", duration: 4000), spin: (keyframes: ..., duration: 6000))
```

Bouncing balls



```
anim: (ball0: (keyframes: (...), duration: 1500, delay: 0), ball1: ...)
```

Pendulums



```
mark("double", ..poses.map(pose)) + anim: (double: (duration: 8000, direction: "alternate"))
```

Waves



```
mark("sea", ..range(0, 25).map(sea)) + anim: (sea: (duration: 4000))
```

Liu Hui's π

$$6 \cdot 12 \cdot 24 \cdot 48 \cdot 96$$



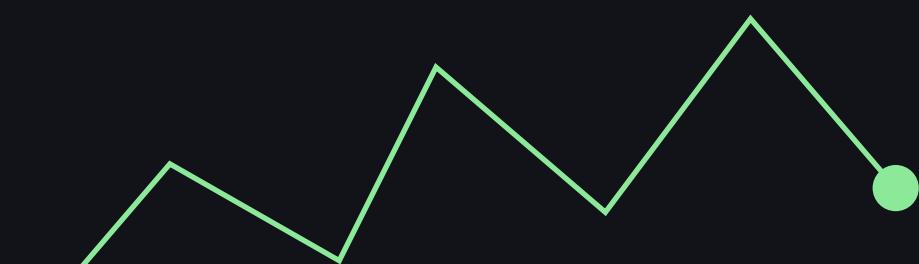
$$\pi \approx 3.1410$$

```
mark("poly", ..range(0, 5).map(exhaust))
```


Growing shapes

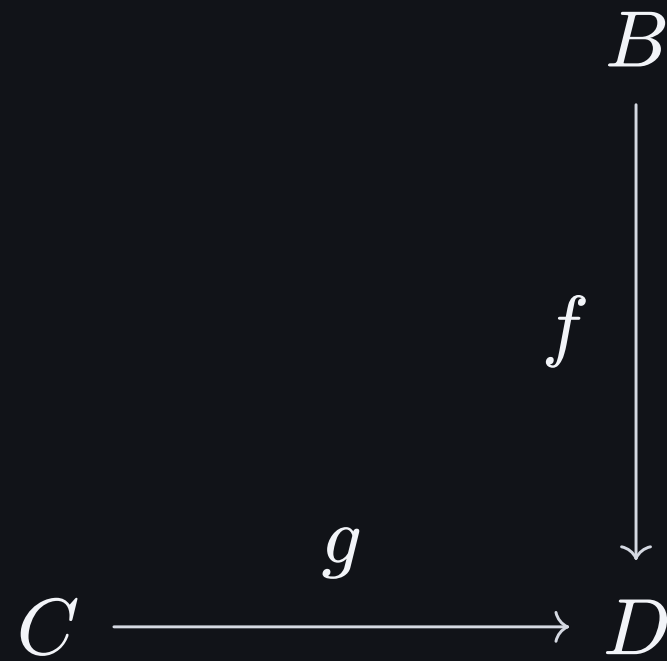


8 sides

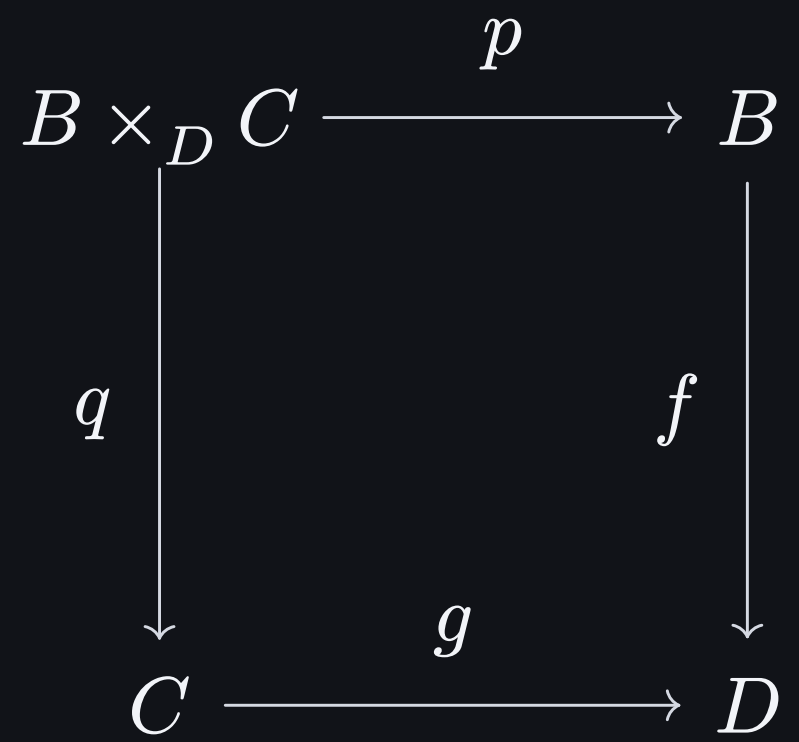


```
mark("gon", ..range(3, 9).map(gon)) · mark("trail", ..range(0, 6).map(trail)) · mark("bend", ..range(0, 3).map(bend))
```

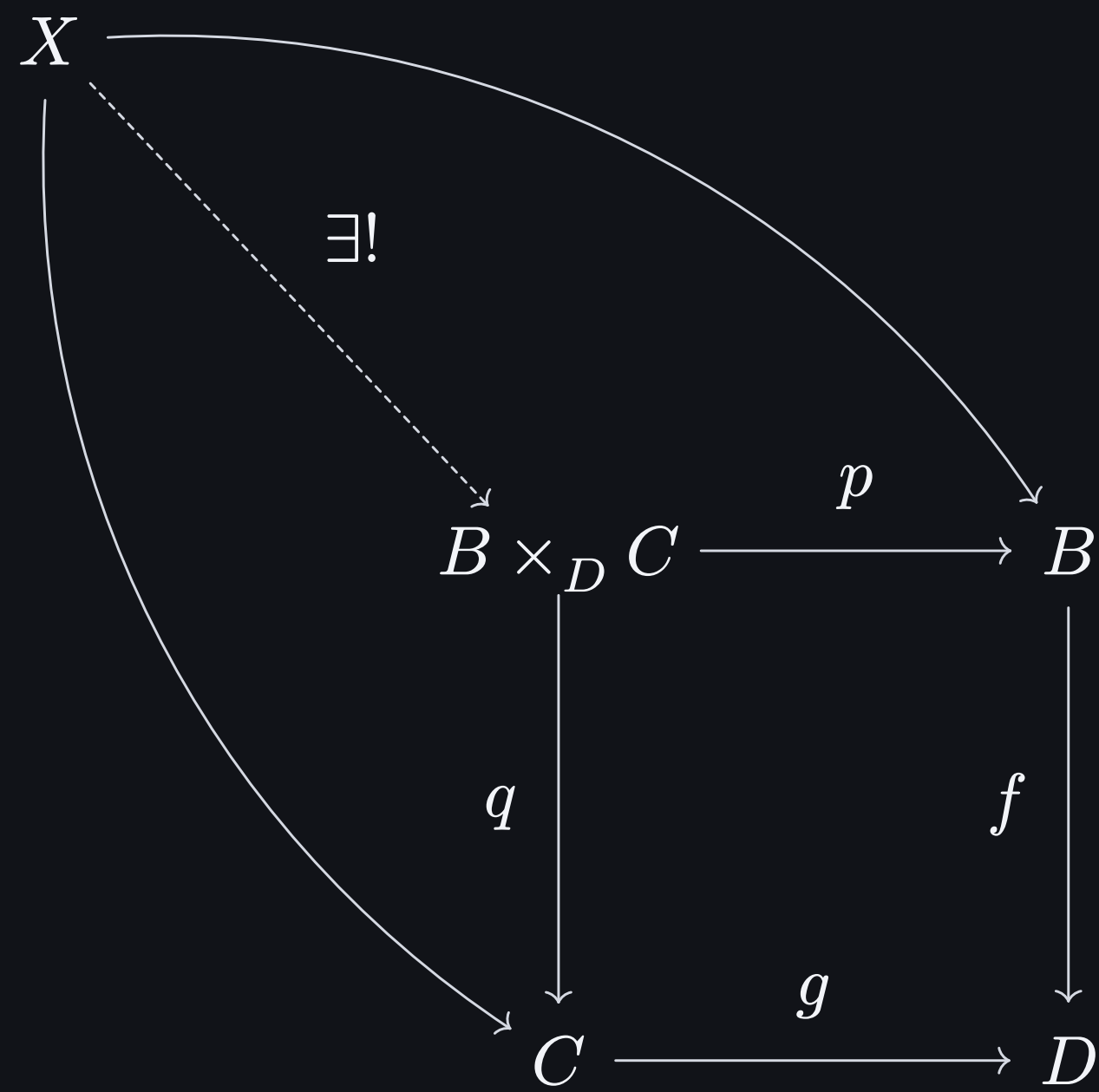
Pullback



Pullback

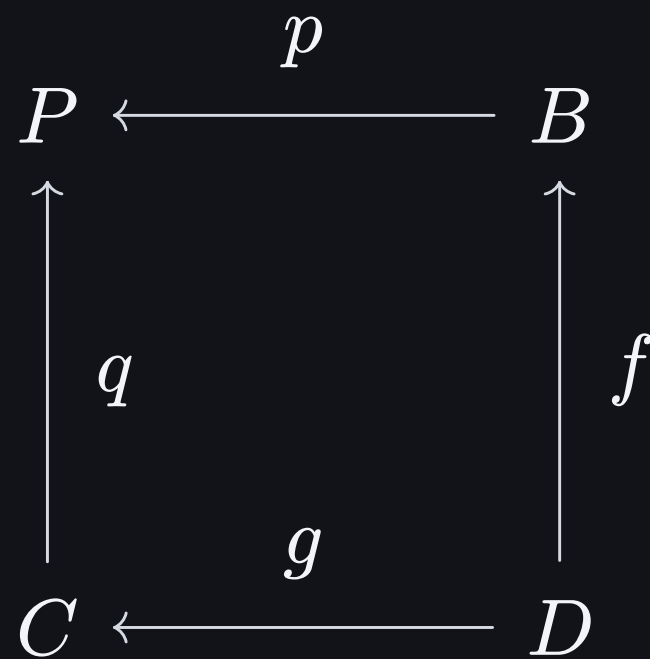


Pullback



```
node((0, 0), mark("pbA", transition: "zoom")[$B \times_D C$])
```

Duality



```
mark("dual", duo(false), duo(true))
```